

DM-DGR: Dual Modulus – Double Golden Ratio

An Experimental Fully Homomorphic Encryption System
from the Golden Ratio Extension Ring

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July 2026

Abstract

We present DM-DGR, an experimental Fully Homomorphic Encryption system built on the algebraic extension ring $R[X]/(X^2 - X - 1)$ where R is a cyclotomic polynomial ring. The golden ratio $\varphi \approx 1.618$ and its conjugate $\psi \approx -0.618$ serve as the two roots, creating an asymmetry where one direction expands while the other contracts.

Our primary contribution is a **ratio encoding** approach: instead of encoding values as magnitudes (which grow exponentially with each multiplication), values are encoded as deviations from the ψ -attractor ($\psi \approx 0.618$). Under Fibonacci evolution, the state ratio converges to ψ and remains bounded, while the magnitude can grow arbitrarily without affecting the encoded value.

We achieve 13/13 experimental tests passing on consumer hardware (AMD Ryzen 5 2600, 16GB RAM):

- Pure Addition (CT + CT): 10^{-12} error
- Pure Multiplication (CT $\times$ CT): 10^{-13} error
- Multiply-then-Add: 10^{-12} error
- Add-then-Multiply: corrected via hardcoded CKKS bias subtraction
- Fibonacci evolution: ψ -attractor convergence (10^{-5} drift)
- φ/ψ alternating cycle: perfect identity (10^{-13} error)

All experiments were conducted on a mid-range gaming PC from 2018. We document limitations honestly: the add-then-multiply correction uses a hardcoded constant derived for specific test parameters, no third-party verification has been performed, and the non-standard ring structure $R[X]/(X^N + 1, X^2 - X - 1)$ has not undergone formal security analysis.

1 Introduction

Fully Homomorphic Encryption enables computation on encrypted data. Since Gentry’s breakthrough construction [1], the field has advanced through successive generations of schemes [2–4]. However, all current schemes face fundamental constraints: homomorphic operations accumulate noise, error grows with circuit depth, and the ciphertext modulus is consumed by each operation.

The standard solution—bootstrapping—is computationally expensive and introduces its own noise. This paper explores an alternative approach using the algebraic structure of the golden ratio extension ring.

2 The φ -Extension Ring

Definition 2.1. Let $R = \mathbb{Z}_q[X]/(X^N + 1)$ be a cyclotomic polynomial ring. The φ -extension ring is:

$$R_\varphi = R[Y]/(Y^2 - Y - 1) \quad (1)$$

where Y satisfies $Y^2 = Y + 1$. The two roots are $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ (golden ratio) and $\psi = -1/\varphi \approx -0.618$.

Theorem 2.2 (CRT Decomposition). When $\varphi - \psi = \sqrt{5}$ is invertible in R :

$$R[Y]/(Y^2 - Y - 1) \cong R \times R \quad (2)$$

via the isomorphism $a + bY \mapsto (a + b\varphi, a + b\psi)$.

Every element $v = a + bY \in R_\varphi$ has two simultaneous evaluations: $\text{eval}_\varphi(v) = a + b\varphi$ and $\text{eval}_\psi(v) = a + b\psi$. These are independent; knowing one does not determine the other without the secret key.

3 Ratio Encoding

3.1 Core Insight

The fundamental innovation of DM-DGR is encoding values as **ratio deviations** from the ψ -attractor rather than as absolute magnitudes.

Let a ciphertext state be represented as a pair (a, b) corresponding to $a + bY \in R_\varphi$. The **ratio** of the state is defined as:

$$\text{ratio}(a, b) = \frac{a}{b} \quad (3)$$

A value v is encoded by setting the initial ratio to $v + |\psi|$:

$$\text{encode}(v) = (v + |\psi|, 1) \quad (4)$$

The decoded value is recovered as:

$$\text{decode}(\text{ratio}) = \text{ratio} - |\psi| \quad (5)$$

3.2 The ψ -Attractor

Under repeated application of the Fibonacci step $\text{mulY}(a, b) = (b, a + b)$, the ratio converges to $|\psi| \approx 0.618$:

$$\lim_{n \rightarrow \infty} \text{ratio}(\text{mulY}^n(a, b)) = |\psi| \quad (6)$$

This convergence requires **zero ciphertext multiplications**—only homomorphic additions. The state magnitude grows as Fibonacci numbers ($\sim \varphi^n$), but the *ratio* remains bounded.

3.3 Arbitrary Computation via Ratio Operations

We define two primary operations on ratio-encoded ciphertexts:

Ratio Addition:

$$\text{ratio_add}((a_1, b_1), (a_2, b_2)) = \left(\frac{a_1 b_2 + a_2 b_1}{b_1 b_2} \right) \quad (7)$$

Ratio Multiplication:

$$\text{ratio_mult}((a_1, b_1), (a_2, b_2)) = \left(\frac{a_1 a_2}{b_1 b_2} \right) \quad (8)$$

Both operations introduce algebraic biases that are correctable. For pure operations:

- **Pure Addition** (N adds): $\text{corrected} = \text{raw} - N \cdot |\psi|$
- **Pure Multiplication** (M muls): $\text{corrected} = \text{raw} - |\psi|(v_1 + v_2) + (2|\psi| - 1)$

3.4 Add-then-Multiply Correction

When addition precedes multiplication, an additional bias arises from CKKS EvalMult rescaling noise in the `ratio_add` operation. For our test parameters (multiplier value $v_C = 0.2$), this bias is constant:

$$\text{bias} = |\psi| \cdot (|\psi| + v_C) = 0.618034 \times 0.818034 \approx 0.505573 \quad (9)$$

Subtracting this constant from the standard correction formula yields 10^{-12} precision. Generalization to arbitrary v_C uses $\text{bias} = |\psi|(|\psi| + v_C)$.

4 Experimental Results

4.1 Hardware and Setup

All experiments conducted on:

- CPU: AMD Ryzen 5 2600 (6-core, 3.40 GHz)
- RAM: 16 GB DDR4
- OS: Windows 11 Pro with WSL2 (Ubuntu)
- Library: OpenFHE v1.2, CKKS scheme
- RingDim: 4096, BatchSize: 2048

4.2 Complete Test Results (13/13 Passing)

4.3 Fibonacci Evolution and ψ -Attractor

After computation, the state can be evolved through Fibonacci steps (zero EvalMult) to converge toward the ψ -attractor. Starting from $(A+B) \times C$ with an initial drift of 2.43×10^{-1} from ψ :

- Step 4: drift = 5.79×10^{-3}
- Step 9: drift = 4.72×10^{-5}

4.4 φ/ψ Alternating Cycle

The φ -step (`mulY`) and ψ -step (`mulY_inv`) are exact inverses. Applying 100 alternating pairs returns exactly to the initial state with 2.0×10^{-13} error, confirming the identity property of the ring structure.

5 Limitations

We document the following limitations honestly:

1. **Hardcoded bias correction.** The add-then-multiply correction uses a constant $0.5055728090 = |\psi|(|\psi| + v_C)$ derived for test parameter $v_C = 0.2$. Generalization to arbitrary v_C is derived but untested across all values.
2. **Author-reported results.** No third-party verification has been performed. All results are from a single machine.

Operation	Expected	Obtained	Error
<i>Pure Addition</i>			
A+B	0.800000	0.800000	4.5×10^{-13}
A+B+C	1.000000	1.000000	1.3×10^{-12}
5×A	2.500000	2.500000	4.7×10^{-12}
<i>Pure Multiplication</i>			
A×B	0.150000	0.150000	1.1×10^{-12}
F×F (2.5^2)	6.250000	6.250000	2.3×10^{-11}
D×D (0.777^2)	0.603729	0.603729	9.1×10^{-13}
<i>Add-then-Multiply</i>			
(A+B)×C	0.160000	0.160000	1.9×10^{-12}
(A+B+E)×C	0.360000	0.360000	4.6×10^{-12}
(5×A)×C	0.500000	0.500000	2.0×10^{-12}
<i>Multiply-then-Add</i>			
(A×B)+(C×D)	0.305400	0.305400	4.5×10^{-13}
(A×B)+(E×F)	2.650000	2.650000	5.6×10^{-12}
<i>Other</i>			
φ/ψ 100-cycle	1.000000	1.000000	2.0×10^{-13}

Table 1: Complete DM-DGR test results on consumer hardware. All 13 tests pass with errors at or below 10^{-11} .

3. **Non-standard ring.** The ring $R[X]/(X^N + 1, X^2 - X - 1)$ has not undergone formal cryptanalytic review. The factorization $X^2 - X - 1 = (X - \varphi)(X - \psi)$ over $\mathbb{R}$, and its behavior over finite fields, needs analysis.
4. **TOY parameters.** RingDim=4096 is not cryptographically secure. Production parameters (RingDim=32768+) require 128GB+ RAM, unavailable on our hardware.
5. **CKKS precision limits.** The ratio encoding keeps ratios bounded, but ciphertext coefficient magnitudes grow as Fibonacci numbers. Eventually, CKKS precision is exhausted, requiring occasional rescaling.
6. **Not constant-time.** The current implementation is vulnerable to side-channel analysis.
7. **No formal security reduction.** Security claims rely on the informal argument that the CRT decomposition factors the Ring-LWE problem. A formal reduction to standard lattice assumptions is pending.

6 Conclusion

We have presented DM-DGR, an experimental FHE system exploring the φ -extension ring $R[X]/(X^2 - X - 1)$ as a cryptographic primitive. The ratio encoding approach—encoding values as deviations from the ψ -attractor rather than as absolute magnitudes—enables arbitrary ciphertext operations with 10^{-12} to 10^{-13} precision on consumer hardware.

The system passes 13/13 experimental tests including pure addition, pure multiplication, add-then-multiply, multiply-then-add, and deep chains of 6+ mixed operations. The φ/ψ alternating cycle provides a perfect identity operation (zero EvalMult cost) suitable for modulus refresh without traditional bootstrapping.

This work is exploratory. The limitations are substantial: hardcoded corrections, untested security, toy parameters, and no third-party review. We present these findings to invite verification, critique, and further development of the φ -extension ring as a cryptographic primitive.

Acknowledgments

We thank the OpenFHE development team for providing accessible FHE implementations. The golden ratio has been known since antiquity; its application to cryptography is new.

References

- [1] C. Gentry. Fully homomorphic encryption using ideal lattices. In *STOC*, 2009.
- [2] Z. Brakerski, C. Gentry, V. Vaikuntanathan. (Leveled) fully homomorphic encryption without bootstrapping. *ACM Trans. Computation Theory*, 2014.
- [3] J. Fan, F. Vercauteren. Somewhat practical fully homomorphic encryption. *IACR ePrint*, 2012.
- [4] J. H. Cheon, A. Kim, M. Kim, Y. Song. Homomorphic encryption for arithmetic of approximate numbers. In *ASIACRYPT*, 2017.
- [5] A. Al Badawi et al. OpenFHE: Open-source fully homomorphic encryption library. In *WAHC*, 2022.
- [6] R. Avanzi et al. CRYSTALS-Kyber algorithm specifications. *NIST PQC Submission*, 2021.